Similar Triangles: Sides, Perimeter, and Area
Answer Key

High-School Geometry   Similarity

Quick Answers

1. 2.5	4. 90	7. 8	10. 1.5, 108
2. 30	5. 8	8. 20	
3. 32	6. 39	9. 9.8	

Curriculum

Level: High-School Geometry   Similarity
HSG-GMD.A.1–A.3 – *problems 4, 8, 10*
HSG-SRT.A.1–A.3, HSG-SRT.B.4–B.5 – *problems 1, 2, 3, 5, 6, 7, 9*
Difficulty 1–5 of 5 across 13 tagged checks.

Common wrong answers

4. *If they got 60: scaled the area by the side ratio instead of by its square*
8. *If they got 26.67: scaled the perimeter by the area ratio instead of by its square root*

1. $\triangle ABC \sim \triangle DEF$, $AB = 6$ cm, $DE = 15$ cm — scale factor.

Step 1. The scale factor from $\triangle ABC$ to $\triangle DEF$ is the ratio of a side of DEF to its corresponding side of ABC, in that order.

Step 2. $k = \dfrac{DE}{AB} = \dfrac{15}{6} = \dfrac{5}{2}$.

$$\boxed{k = 2.5}$$

Listen for: $\frac{6}{15} = 0.4$, which is the factor the other way round. The direction is part of the answer.

2. $\triangle ABC \sim \triangle PQR$: $AB = 8$, $BC = 12$, $PQ = 20$ — find QR.

Step 1. Corresponding sides come from the similarity statement: $AB \leftrightarrow PQ$ and $BC \leftrightarrow QR$, so $\dfrac{AB}{PQ} = \dfrac{BC}{QR}$.

Step 2. $\dfrac{8}{20} = \dfrac{12}{QR}$, so $8 \cdot QR = 12 \cdot 20 = 240$.

Step 3. $QR = \dfrac{240}{8} = 30$.

$$\boxed{QR = 30 \text{ cm}}$$

Check: $k = \frac{20}{8} = 2.5$ and $12 \times 2.5 = 30$ ✓

3. Garden beds: sides 9 cm and 12 cm, smaller perimeter 24 cm.

Step 1. Perimeter is a sum of sides, so it scales by the same factor the sides do — no squaring here.
Step 2. $k = \frac{12}{9} = \frac{4}{3}$.
Step 3. $P_{\text{large}} = 24 \cdot \frac{4}{3} = 32$.

$$\boxed{32 \text{ cm}}$$

Teaching note: a student can also check by scaling each side and adding — slower, but it shows why the shortcut works.

4. Tiles: sides 10 cm and 15 cm, smaller area 40 cm^2.

Step 1. Area is measured in two dimensions, so it scales by k^2, not by k.
Step 2. $k = \frac{15}{10} = \frac{3}{2}$, so $k^2 = \frac{9}{4} = 2.25$.
Step 3. $A_{\text{large}} = 40 \times 2.25 = 90$.

$$\boxed{90 \text{ cm}^2}$$

Common wrong answer: 60 — the area was multiplied by $\frac{3}{2}$ instead of by $\left(\frac{3}{2}\right)^2$.

5. $\overline{DE} \parallel \overline{BC}$, $AD = 6$, $AB = 15$, $BC = 20$.

Step 1. (a) Because $\overline{DE} \parallel \overline{BC}$, the transversal $\overline{AB}$ makes $\angle ADE \cong \angle ABC$ (corresponding angles), and $\angle A$ is shared. Two pairs of congruent angles is the **AA** criterion, so $\triangle ADE \sim \triangle ABC$.
Step 2. (b) Corresponding sides then satisfy $\frac{AD}{AB} = \frac{DE}{BC}$, i.e. $\frac{6}{15} = \frac{DE}{20}$.
Step 3. $15 \cdot DE = 6 \cdot 20 = 120$, so $DE = 8$.

$$\boxed{DE = 8 \text{ cm}}$$

Manual review: part (a) is the written half. Accept any correct AA argument; the shared angle at A must be named, not assumed.

6. Flagpole: student 6 ft with a 4 ft shadow, pole shadow 26 ft.

Step 1. The sun's rays strike both objects at the same angle, and both stand vertically, so the two triangles are similar by AA — height to shadow is the pair to compare.
Step 2. $\frac{\text{height}}{\text{shadow}}$ is the same for both: $\frac{6}{4} = \frac{h}{26}$.
Step 3. $4h = 6 \cdot 26 = 156$, so $h = 39$.

$$\boxed{h = 39 \text{ ft}}$$

Sanity check: the pole's shadow is 6.5 times the student's, and $6 \times 6.5 = 39$ ✓

7. Perimeters 45 cm and 30 cm, $AB = 12$ cm — find XY.

Step 1. The perimeter ratio IS the side ratio, so the two perimeters give k without knowing a single side of $\triangle XYZ$.
Step 2. $k = \frac{30}{45} = \frac{2}{3}$ (from ABC to XYZ).
Step 3. $XY = 12 \cdot \frac{2}{3} = 8$.

$$\boxed{XY = 8 \text{ cm}}$$

Listen for: 18 — from scaling by $\frac{3}{2}$, the wrong direction.

8. Areas 27 cm^2 and 48 cm^2, smaller perimeter 15 cm.

Step 1. The area ratio is k^2, and the perimeter needs k — so take a square root before scaling anything.

Step 2. $k^2 = \dfrac{48}{27} = \dfrac{16}{9}$, so $k = \sqrt{\dfrac{16}{9}} = \dfrac{4}{3}$.

Step 3. $P_{\text{large}} = 15 \cdot \dfrac{4}{3} = 20$.

$\boxed{20 \text{ cm}}$

Common wrong answer: 26.67 — the perimeter was scaled by the AREA ratio $\frac{16}{9}$ instead of by its square root.

9. Perimeters 30 cm and 42 cm, a 7 cm side.

Step 1. Perimeters scale like sides, so their ratio is the factor that carries the 7 cm side across.

Step 2. $k = \dfrac{42}{30} = \dfrac{7}{5} = 1.4$.

Step 3. Corresponding side $= 7 \times 1.4 = 9.8$.

$\boxed{9.8 \text{ cm}}$

Check: 9.8 is a little under one and a half times 7, which matches a perimeter that grew from 30 to 42 ✓

10. Challenge: perimeters 36 cm and 54 cm, area of $\triangle ABC$ is 48 cm^2.

Step 1. (a) Perimeter ratio is the scale factor: $k = \dfrac{54}{36} = \dfrac{3}{2}$.

$\boxed{k = 1.5}$

Step 2. (b) Area scales by $k^2 = \left(\dfrac{3}{2}\right)^2 = \dfrac{9}{4} = 2.25$.

Step 3. $A_{DEF} = 48 \times 2.25 = 108$.

$\boxed{108 \text{ cm}^2}$

Step 4. (c) *Model answer:* a triangle's area is $\frac{1}{2}bh$. Doubling every side doubles the base and doubles the height, and the area depends on both, so it grows by $2 \times 2 = 4$. Scaling a length stretches one direction; scaling an area stretches two, which is exactly why the area ratio is k^2 while the perimeter ratio is k.

Manual review: part (c) is the written half — look for both dimensions being named, not just the word "squared".